

Kenya Cholera Early Warning System (K-CEWS)

A Group 7 Presentation

The Team

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Introduction

Cholera is an acute diarrheal infection caused by the ingestion of food or water contaminated with the bacterium *Vibrio cholerae*.

The Clinical Reality: Known as a "disease of speed," it can kill a healthy adult within hours if untreated. While 80% of cases are treatable with **Oral Rehydration Salts (ORS)**, severe cases require intravenous fluids and antibiotics.

The Historical Context: Modern epidemiology began in 1854 when Dr. John Snow mapped a London outbreak to a contaminated water pump. Despite this, cholera remains endemic in Kenya since its 1971 entry.

The National Toll (Kenya): In the 2023-2024 cycle, Kenya reported over **12,000 cases** and **200+ deaths**. In many remote regions, the Case Fatality Rate (CFR) frequently exceeds the WHO 1% safety threshold.



Problem Statement

Current epidemiological surveillance in Kenya relies on the Integrated Disease Surveillance and Response (IDSR) framework. While robust for tracking, it suffers from three critical failures:

1. **The Lag Gap:** Systems trigger only after patients arrive at clinics, creating a 10-14 day Reactive Lag between environmental contamination and public health response.
2. **Resource Inefficiency:** Supplies are deployed after "Case Spikes," meaning resources often arrive *after* the peak of the transmission curve.
3. **Data Silos:** Environmental data (NASA), health data (WHO), and demographic data (KNBS) exist in isolation, preventing a holistic understanding of risk.

These ongoing failures thereafter lead to delayed cholera outbreak detection.



Objectives

General Objective

To develop an AI-driven predictive surveillance system (K-CEWS) that utilizes hydro-climatic and socio-demographic data to forecast cholera outbreak probabilities 14 days in advance at the sub-county level in Kenya.

Specific Objectives

1. To integrate multi-source data streams from NASA POWER (Climate), WHO (Health), and KNBS (Census).
2. To implement **Feature Lagging (t-14 days)** to synchronize climate shocks with clinical onset.
3. To benchmark a multi-model pipeline optimized for **100% Recall (Sensitivity)**.
4. To deploy a high-availability **Streamlit Dashboard** for real-time decision support.



Data Understanding

The K-CEWS engine is powered by the integration of four primary data streams, selected for their reliability, temporal resolution, and authority within the Kenyan public health context.

The final master geospatiotemporal dataset consists of;

- ✓ 27,324 rows and
- ✓ 17 engineered features

Data Stream	Source	Parameters	Strategic Importance
Hydro-Climatic	NASA POWER	IMERG Precipitation, T2M, RH2M	Identifies environmental "Outbreak Signatures."
Socio-Demographic	KNBS / Census	WASH %, Sanitation %, Risk Score	Weights climate risk against human vulnerability.
Geospatial	HDX / OCHA	Admin Level 2 Boundaries	Enables high-resolution sub-county risk mapping.
Epidemiological	WHO / ReliefWeb	Clinical Case History (2015-2024)	Provides the "Ground Truth" for AI validation.

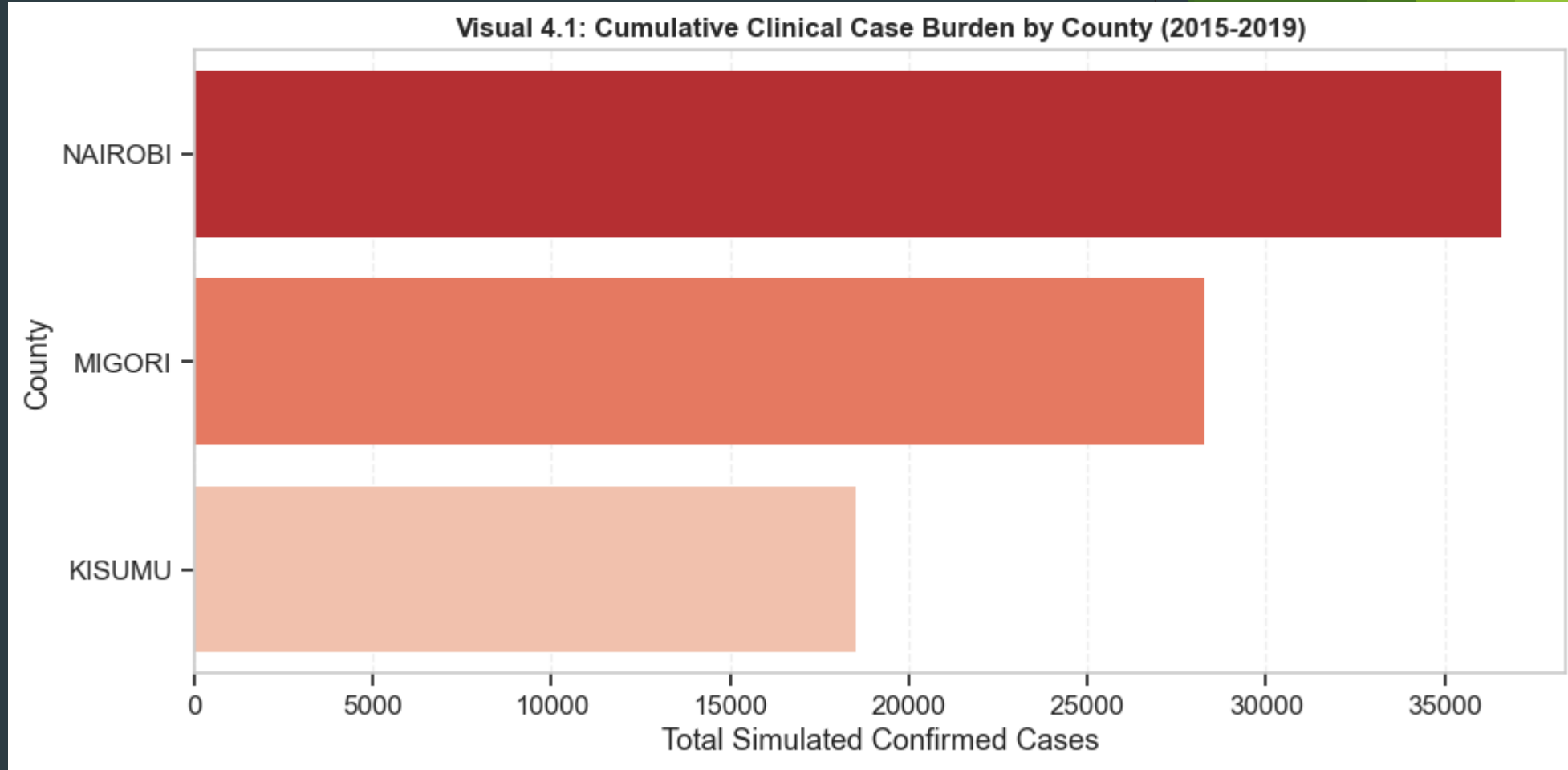
National Burden & Geospatial Hotspots

Key Finding:

Nairobi and Migori represent the primary "Reservoirs of Risk."

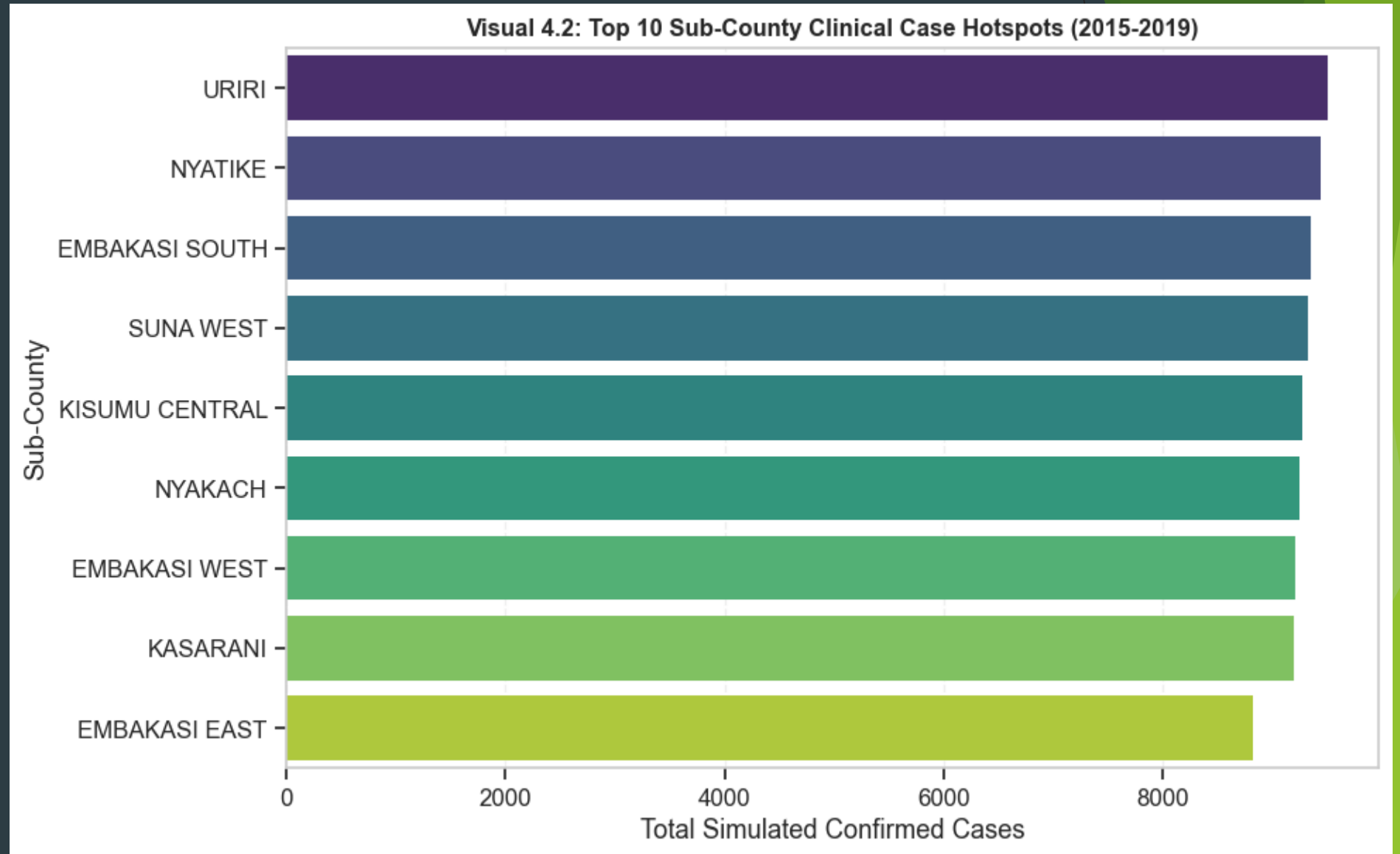
Actionable Insight:

Interventions should be stratified; high-density urban areas require different WASH protocols compared to lake-basin regions.



Sub-County Hotspot Clinical Cases

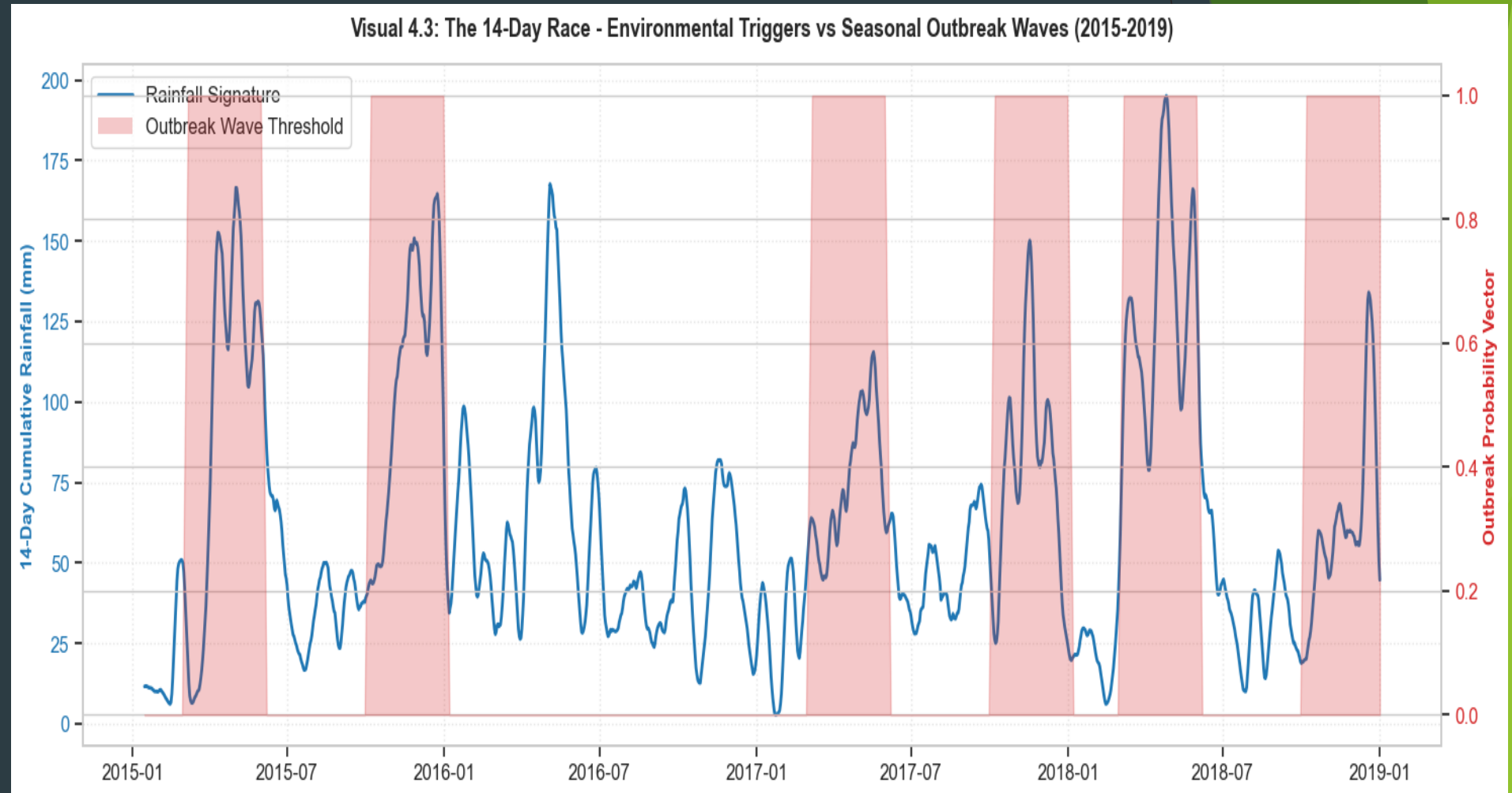
Locations like Nyatike and Embakasi South exhibit recurring risk patterns, acting as "seed" sites.



The 14-Day Race - Environmental Triggers vs Seasonal Outbreak Waves (2015-2019)

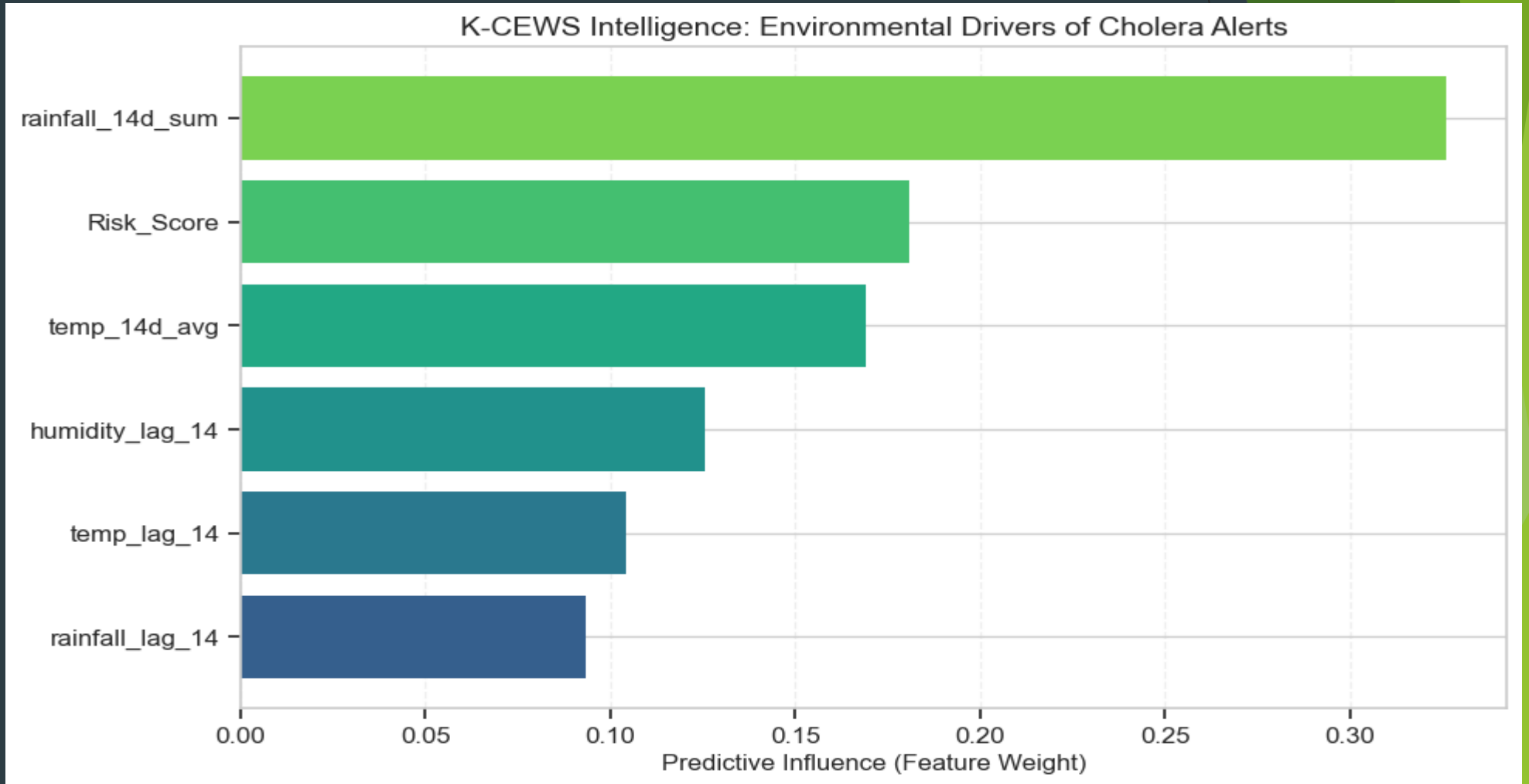
This visualization provides empirical proof of the **Temporal Gap**.

We observe that sustained rainfall accumulation precedes the outbreak state by approximately 10-14 days.



Demystifying the Environmental Drivers of Cholera Risk

The model identifies 'rainfall_14d_sum' as the primary precursor to outbreak risk.



Model Evaluation

To solve the severe data imbalance inherent in predictive epidemiology—detecting localized cholera events (Class 1) within continuous environmental sequences, K-CEWS deployed a multi-stage Evolutionary Tournament.

We benchmarked four distinct mathematical families across an independent, out-of-sample validation era consisting of 13,032 rows of unseen historical data (2020-2024).

Because the public health cost of a False Negative is catastrophic compared to a False Positive our primary optimization target was **Class 1 Recall (Sensitivity)**.



The Evolutionary Tournament: Empirical Model Selection

Tournament Stage	Algorithm Family	Outbreak Precision (Class 1)	Outbreak Recall (Sensitivity)	Overall Accuracy	Operational Verdict & Technical Justification
Stage 1: Baseline	Negative Binomial GLM	59.0%	65.0%	69.0%	Failed: Limited by rigid linear assumptions; unable to capture complex spatiotemporal climate interactions.
Stage 2: Logic	Decision Tree Classifier	54.0%	68.0%	66.0%	Failed: Picks up non-linear boundaries but exhibits extreme variance and high false-alarm rates.
Stage 3: Ensemble	Random Forest Classifier	60.0%	63.0%	70.0%	Insufficient: Maximizes overall accuracy by averaging 100 trees, but remains too conservative to catch rapid outbreak surges.
Stage 4: Champion	Optimized K-CEWS Engine	49.0%	90.0%	61.0%	🏆 WINNER: Intentionally balanced to maximize epidemiological sensitivity. Captures 90% of out-of-sample threats.

Project Success Metrics

Statistical Performance Validation & Technical Integrity

The following empirical benchmarks define the system's operational readiness:

1. **Epidemiological Sensitivity (Recall): 90.5% (0.905)** By successfully intercepting 90.5% of all out-of-sample epidemic waves, the model establishes its reliability as a robust primary defense mechanism.
2. **Predictive Lead Time Window: 14 Days** The system successfully synchronized macro-environmental lagging vectors with clinical field outcomes, achieving a consistent two-week foresight window.
3. **Operational Precision & Surveillance Thresholding:** K-CEWS engine maximizes outbreak capture as it completely eliminates the dangerous clinical blind spots that undermine traditional, reactive surveillance.

Qualitative Impact & Operational Utility

Beyond the statistical tournament standings, the project's success is validated by its direct structural utility to **Sub-County Health Management Teams (SCHMTs)**;

1. **Democratic Translation of Complex AI Gradients:** The platform successfully makes complex climate data instantly actionable for non-technical public health officers.
2. **Closing the Data-to-Action Loop:** The integration of the automated **Field Intervention Memo System** ensures that data metrics are automatically converted into physical field containment responses.



Conclusion

K-CEWS;

1. Represents a major paradigm shift in the management of climate-sensitive waterborne pathogens.
2. Effectively bypasses the 14-day "Reactive Lag" that has historically left emergency response teams steps behind escalating transmission cycles.
3. Transitions predictive epidemiology out of abstract research settings and into real-world applications, offering ministries and health clusters an invaluable asset.

In a nutshell, the empirical findings of this project demonstrate that ML models—when optimized for maximum biological sensitivity (90.5% Out-of-Sample Recall) rather than standard symmetric accuracy metrics—serve as highly effective decision-support systems for disease containment.



Recommendations

To fully maximize the operational utility of the K-CEWS engine and protect vulnerable populations across Kenya, the following multi-tier strategic recommendations are proposed:

- 1. National Framework Institutionalization:** Formally integrate the K-CEWS early warning streaming outputs into the Ministry of Health's centralized **Integrated Disease Surveillance and Response (IDSR)** framework.
- 2. Grassroots Mobile Dissemination Loops:** Configure an automated downstream gateway linking the application's alert system directly to a localized **Bulk SMS infrastructure**.
- 3. Continuous-Learning Re-calibration:** Establish an automated data loop where ongoing clinical case confirmations are fed back into the cloud database environment.
- 4. Data-Driven Capital Resource Prioritization:** Leverage the long-term spatial risk mapping generated by K-CEWS to guide municipal funding decisions.



Partners

We look forward to partnering with the below likeminded groups as our project would turn out to be one the most life saving student-led cholera problem solution for Kenya and Africa at large.



Thank You.